Graphic Design

Advanced Colour Theory & Print Terminology

Summary Notes

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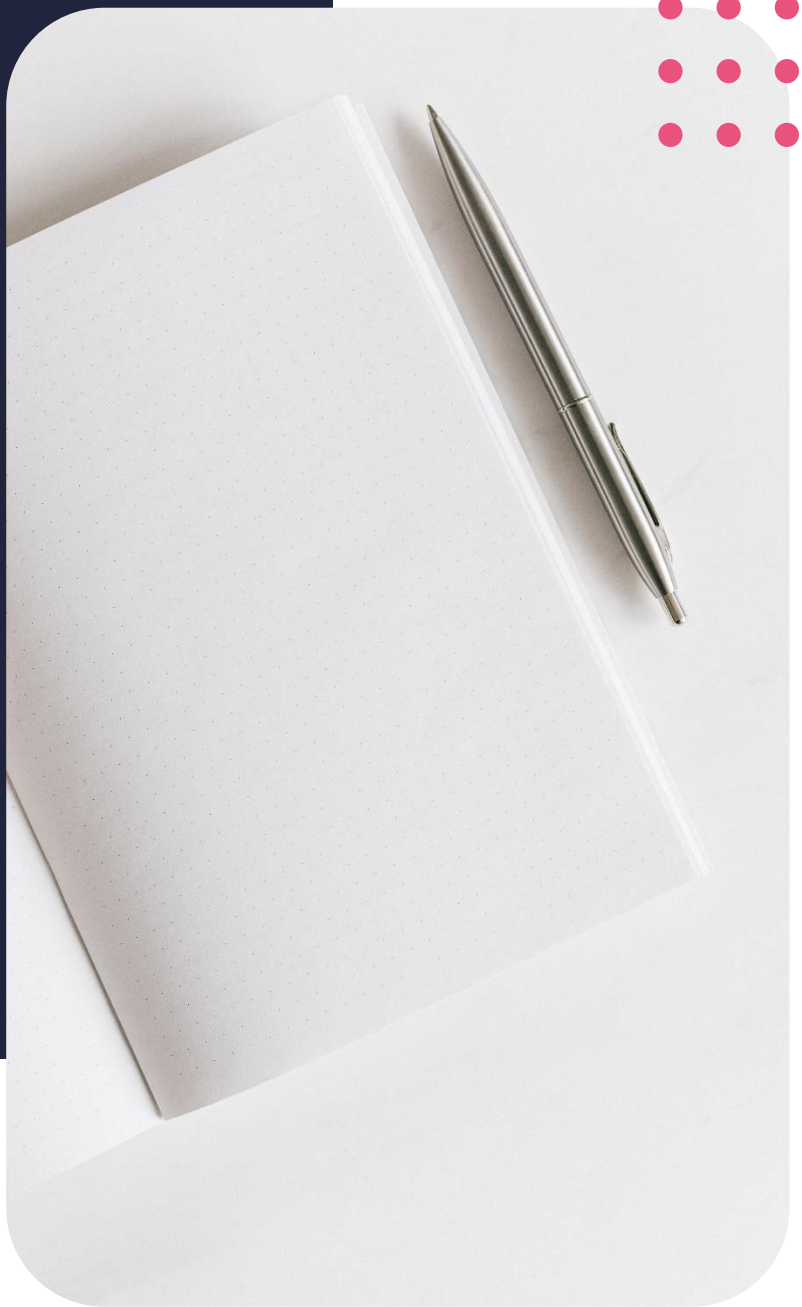


Lesson outcomes

By the end of this lesson, you should be able to:

- Lesson outcome 1: Understanding colour theory
- Lesson outcome 2: Understanding colour models
- Lesson outcome 3: Pantone colour matching systems

Introduction

Have you ever realised that when printing, there is no white ink? Let that sink in a bit…

Why have you not even considered printing with white ink?

When it comes to printing, we print with a SUBTRACTIVE colour model that consists of CMYK. When combined, these colours form black. So, technically, these colours can't form white. Also, considering that paper is white, it would not make sense to have white ink.

However, we do get the occasional project that entails white print because our printing surface is not always white.

In this lesson, I will explain the technique of printing with white, gold and other colours that can't be mixed by using the CMYK colour model. And guess what? Print is not dead! We have amazing printing techniques that will blow your mind!

Advanced Colour Theory

Just as you thought you know all there is to know about working with colour, you get introduced to it yet again. This time however, we will take a more practical approach.

Colour helps to create impact; it organises and evokes emotion and what better way than to understand how to work with colour in order to communicate the intended message?

Colour has such an impact that by simply changing the colour of a clothing item or car for example, it will put us off that item. However, colour might evoke a different emotion in you than it does in the person next to you – especially considering that we have different reactions to colours due to cultural, associational or personal preferences.

By simply changing the hue, value or saturation of a colour, that too can create a different mood or evoke different feelings.

This lesson is all about working with colour and understanding the emotional impact it has on society. Also note that this is a guide to understanding basic colour psychology and that the context around colour and even the surrounding colours can have a different effect.

I would like to do a brief recap on the colour wheel we did in our previous module.

So, by dividing the colour wheel in half between orange and yellow as well as blue and purple, we can determine that the colours on our left have a warm undertone and the colours on our right have a cool undertone.

What type of emotion do warm colours mostly convey? Well, on the positive spectrum we have comfort, love and warmth but these colours can also convey a sense of hostility, aggression and anger. Cool colours can convey a sense of calmness and peace but also sadness, so understanding when and how to use colours in graphic design is extremely important.

Let's have a look at these colours individually by starting with red

Red

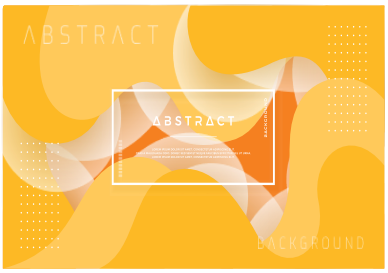
Being a primary colour, red can convey an array of feelings ranging from love and passion to danger and anger.

Understanding the use of colour in different cultures is also an important topic to consider. Red in some Eastern cultures symbolises good fortune and prosperity and is worn by brides on their wedding day. Red can also symbolise revolution and political movements.

When it comes to branding however, red symbolises strength, confidence and power. Red in design terms can be used as an accent colour but when used in its purest form, it can be quite overwhelming and by simply changing the value and saturation of red, it can be quite versatile as well.

Orange

Since orange is a combination of red and yellow, it will take on the properties of both. In branding, orange represents a youthfulness and excitement. It is associated with brands that symbolise enthusiasm and are energising. In nature, a down-toned orange can also symbolise the change in seasons, from autumn to winter. Learn to see the queues colours convey in nature and culture and apply that to your design.

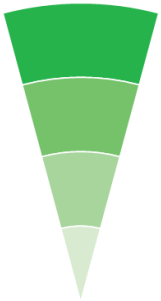


Yellow

Yellow, conveys emotions of happiness, laughter, warmth, optimism, energising, it is attention-grabbing, makes you feel hungry, frustrated or even angry. It can also resemble deceit.

Bright yellows normally lend toward laughter and energy. Light yellows are calm and portray a sense of happiness. Dark yellows like mustard can create the illusion of gold and resemble royalty, wealth or elegance. However, combining yellow with say a white background can be visually disturbing. In branding, yellow is known to grab your audience attention and should therefore be used sparingly.

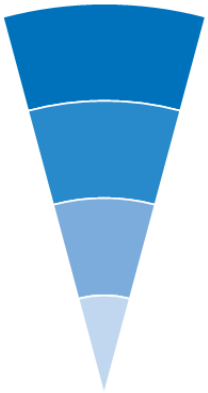
The good news in design however is that by simply adjusting the colour's brightness or saturation, you will be able to make use of it in more ways than you think.



Green

Since green is a combination of blue and yellow, it will take on some of the attributes of both yellow and blue. Brighter greens tend to be more vibrant and energising whereas darker greens tend to be more stable.

In branding, green gives a sense of health and wealth. It can also resemble nature and is often seen in eco-friendly designs.

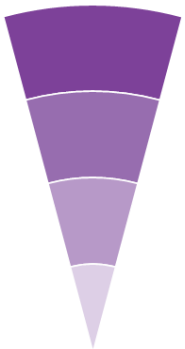


Blue

Moving over to our cool colours, blue is the only primary colour within the cool spectrum. Our cool colours, are more subdued than our warmer colours. Hence, for a cool colour to stand out when combined with a warm colour, you will need to alter the value and saturation of the warm colour. Blue can lower blood pressure and is thus useful to create a calm atmosphere and a relaxing environment. Blue is also a very corporate colour and you will see that a lot of corporates make use of blue in their branding.

Altering the shades of blue can thus then evoke a different emotion. Have you ever wondered where the saying “feeling blue” come from? If you’re not familiar with this saying, to refer to yourself as feeling blue means that you are sad. This term derived from a custom among sailors where if the ship lost its captain during its voyage, she would fly blue flags for that entire journey until returning to her home port.

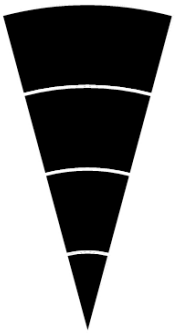
When it comes to branding, blue is one of the most widely used colours and usually communicates wisdom, loyalty and truth. A dark blue is normally used in a very corporate sense since it communicates a sense of professionalism and seriousness.



Purple

Purple being a combination of red and blue gives a sense of royalty and luxury, especially darker purples. Lighter purples tend to resemble creativity and imagination.

Purple can communicate a sense of royalty, luxury, it is sophisticated and creates mystery. Purple also connects to wisdom and spirituality.

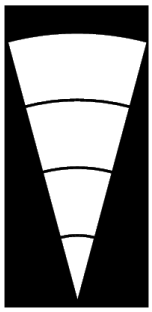


Black

Neutrals are mostly used as a backdrop in design and are normally combined with brighter accent colour. Aid in creating sophistication and neutrals are very much affected by the colours around them.

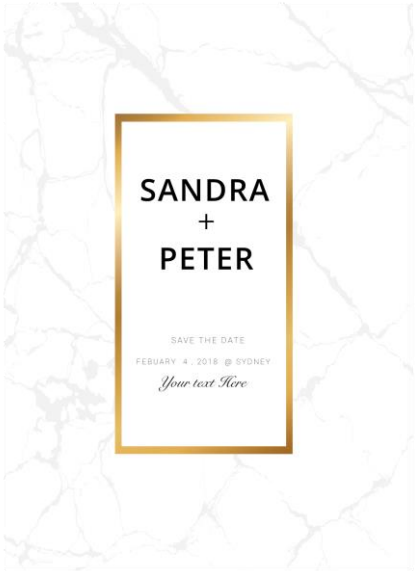
In most cultures, black symbolises death and a sense of mourning.

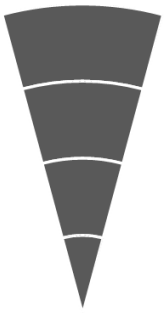
In design however it links to elegance and sophistication. Black is also generally used for body copy and typography.



White

White let's other colours have more impact and create a sense of sophistication. Depending on the backdrop colour but in design, a white logo is rarely seen. You will however find black logos to be converted to white if the background is a dark colour. In order to create contrast of course. White is also a colour that is used in the health care industry and symbolises cleanliness, purity and innocence.





Grey

Grey is probably my favourite colour to use in designs. As with white and black, when it comes to grey in design, it is normally used as a secondary option especially with logos and branding. It is however sometimes used as the primary colour in branding. Mixing shades of grey also creates a timeless, practical approach and when it comes to printing, greyscale is normally very cost effective since you are only making use of one colour.

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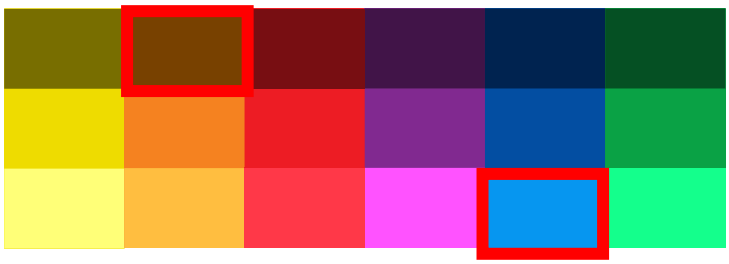
Additional colour terms

Some additional terms we will be making use of a lot, especially when working with the design programs are hue, value and saturation.

Hue refers to the actual colour you will be using. EG, green or blue or yellow. Hues are pure colour.

Value, also known as brightness, refers to how light or dark a colour is. By simply adding either white or black.

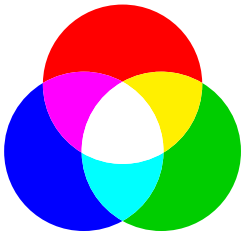


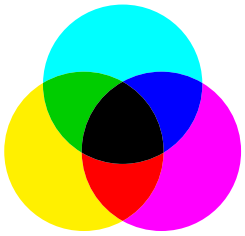
Saturation refers to the intensity of a colour with similar saturation values tend to work well together. Think of it in terms of removing colour and adding more grey. Eventually you will get 100% pure grey when all the colour is removed.

Understanding colour models

Just as you thought you know all there is to know about working with colour, you get introduced to it yet again. This time however, we will take a more practical approach.

Previously you were introduced to the two main colour models: RGB & CMYK. Other colour models you will be familiarised with in this lesson are HSB and Lab colour models.





A colour model is a system for creating a full range of colours from our primary colours.

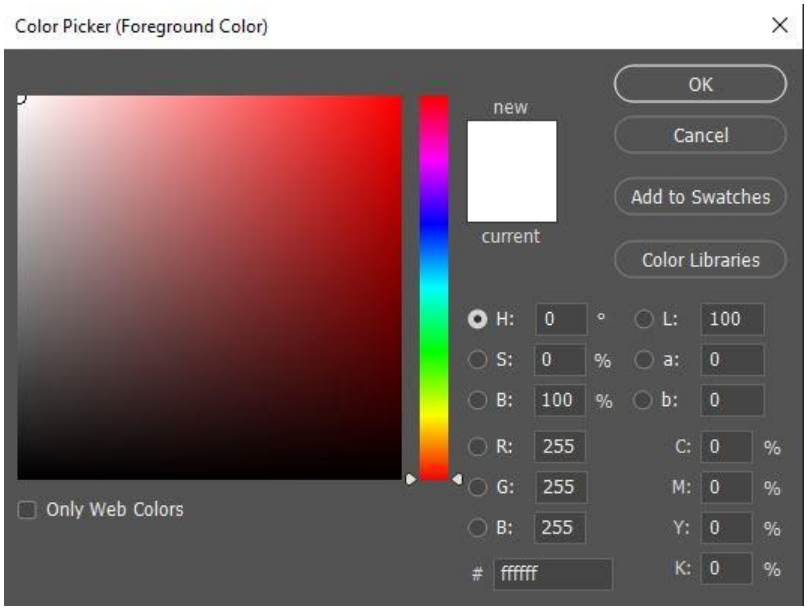
We have additive and subtractive colour models. Additive colour models uses light to display colours and subtractive colour models uses paints or inks. So, RGB is an additive colour model since combining all these colours form white light and CMYK is a subtractive colour model since combining all these colours produces black.

The HSB or HSV colour model is the abbreviation for Hue, Saturation and Brightness (or Value). You can access this colour model on any of the three design programs we have learnt in this course by simply double clicking on the colour palette in your toolbar on the left.

The Lab colour model was introduced by Richard Hurter American colour scientist in the 1940's but only defined by the International Commission on Illumination back in 1976. It expresses colour as three values: L for lightness from black to white; a from green to red and b from blue to yellow. The Lab colour space includes all the colours of the spectrum, even colours outside of our human perception. This is thus the most accurate means of representing colour of all the colour models. It is however not the most used and colours are normally converted to colour models like RGB and CMYK since computers and printers use either three or four colours to represent images. In other words – if we can't even see all the colours of the spectrum, why bother trying to represent them?

Next on our colour picker we have RGB and CMYK, which you are familiar with by now.

The last colour model at the bottom refers to web safe colours and are referred to in hex values. We will get to this topic and designing for web at a later stage.



Pantone colour matching system

The pantone colour matching system is a colour space used in graphic design, printing and other industries including fashion, product design and manufacturing. This colour space was created by Pantone situated in New Jersey.

The pantone colour space is used to print spot colours whereas CMYK is used to print processed colours. So, what is the difference?

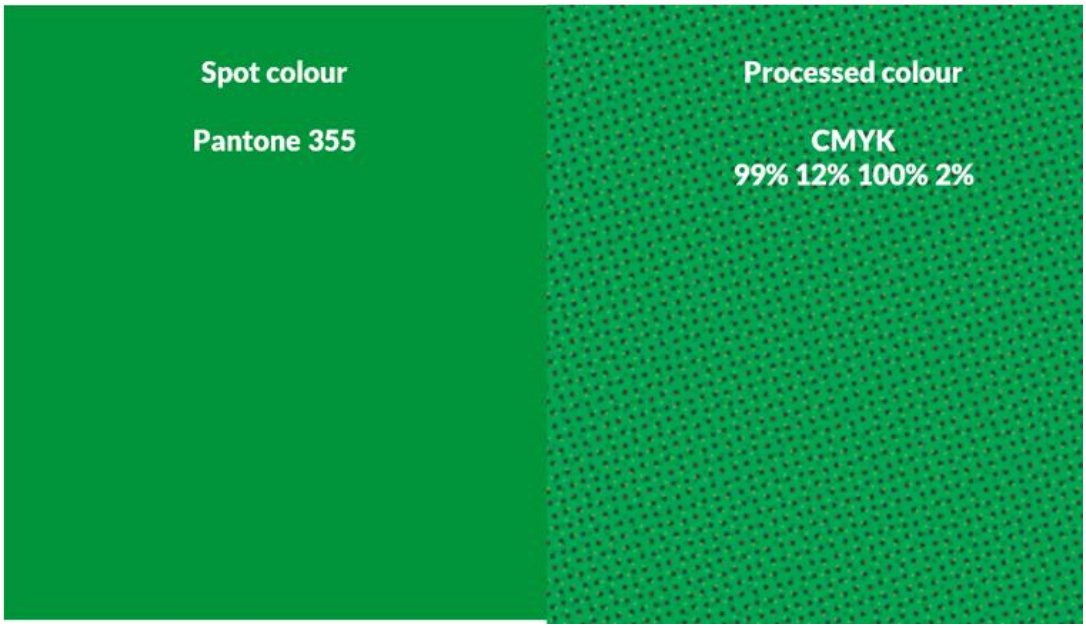
Spot colours are solid inks assigned by a code that aid printers in making sure that the colours are printed accurately. CMYK colours are combined to create a specific colour. In short: a spot colour is a premixed ink of a specific colour and the CMYK colour model uses inks that get mixed in the printing process.

Pantone 109		**CMYK** **1% 16% 100% 0%**
Pantone 355		**CMYK** **99% 12% 100% 2%**
Pantone 2685		**CMYK** **93% 100% 18% 21%**

So, for example: The BP green is coded Pantone 355 and the BP yellow is Pantone 109. The Cadbury purple is Pantone 2685

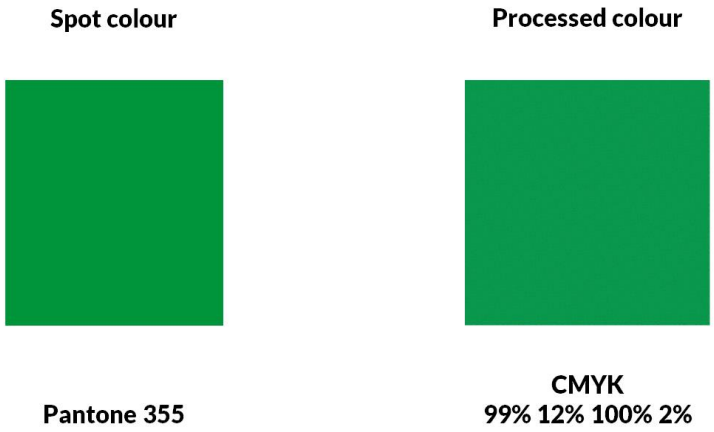
Spot colours come in handy, especially in logos where colours need to be extremely accurate.

Zooming in on a printed document, you can sometimes determine whether the colour is a spot colour or processed colour. By considering what you have just learnt, which colour do you think is a spot, or Pantone colour and which is the processed or CMYK colour?



The colour on the left is the spot colour as you can see that it is printed with one single green colour and the colour on the right is a mixture of CMYK where Cyan and Yellow makes up almost 100% of the ink with 12% magenta added and only 2% black.

When zooming out, you can barely see the difference but with a trained eye, you will soon realise when a colour is a spot or processed colour. You can test this out – next time when you look at a printed item – in extreme close up, use a magnifying glass if need be, determine where spot colours and where processed colours are visible. I think the best example would be to look at a cereal box or magazine cover as not all printed items combine both processed and spot colours.

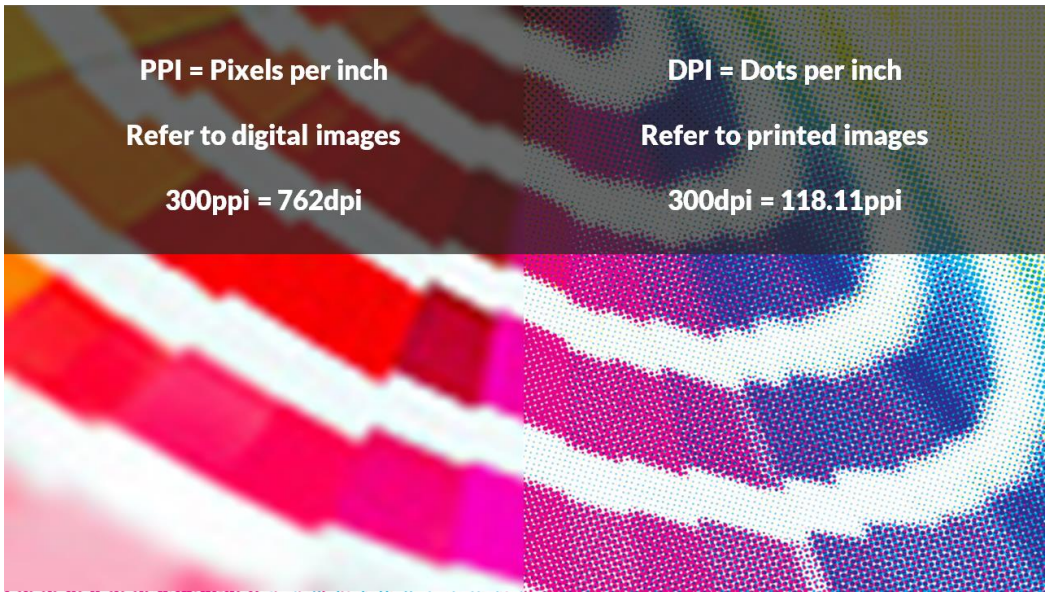


A tip I can give you: like with raster images that consist of millions of pixels and a variety of different coloured squares, so does processed colour also work. So, any printed element that has an array of colours is most likely not a spot colour but a processed colour printed using a halftone screen.

Pixels vs Dots

Pixels and dots are however not the same thing: The brief difference between PPI and DPI is that PPI is used for digital images and DPI is used for printed images.

So, a lot of people get confused with these two terms and use them interchangeably - this is wrong, don't do it!



Getting all technical, we can say that 300ppi is equal to roughly 762dpi and 118,11ppi is equivalent to 300dpi. But don't let this confuse you at the moment – it is just important to note that dpi and ppi are not the same thing and are used interchangeably by many in the wrong way.

Just stick to what you learnt previously – save your images at 300ppi when printing.

So, on the right hand side we have a low resolution digital image and on the left hand side we have a low quality halftone image – the dots are clearly visible here. You can see lower quality halftone processed images clearly in newspapers as the quality printing for newspapers is not has high.

On another note, halftone is simply a term used to break down a processed image into an array of coloured dots in different sizes, like you can see on the right-hand image.

On the right, you can see tiny dots of colour which we refer to as halftone in the printing industry. These tiny dots are a sure sign of a processed image and processed images are measured in dots per inch. DPI refers to the number of printed dots within one inch of an image, almost like pixels per inch you learnt about.

Going back to our spot colours, which we determined is a pre-mixed colour. This then answers our question how to print white? Thinking about it: it is impossible to print white combining the CMYK colour model – let that sink in for a moment…

Well, if we can premix a colour of ink, why not get a white spot colour, or a gold spot colour or a neon pink spot colour! Technically speaking, these colours can't be created using CMYK, so spot colours, and specifically using Pantone spot colours come in quite handy when we need to print colours outside of the CMYK colour model as well.

The only downfall is that it can become quite expensive when you add an additional colour to your already standard four colours, so next time you have a client wanting any other colour that can't be mixed with the standard CMYK colour model, prepare them to pay a lot.

Speaking of printing colours, this brings me to my next topic..

If you are ever privileged enough to possess a Pantone colour swatch, call yourself lucky, or if you feel like spending some money by spoiling a designer friend, buy them a pantone colour swatch – they will love you for it! A pantone colour swatch is a small booklet containing all the different Pantone colours available – it's like buying your fashion-forward friend some Louboutin stilettos.

Printing processes and terms

If printing really interests you, I suggest to visit your local repro house or printing company and see how they operate – this is actually really fascinating!

This lesson will also put things into perspective for you as to why printing is so expensive. Sometimes, the ink itself can be more expensive that gold per gram, so double checking your artwork before printing is crucial.

The printing processes I will be explaining in this lesson are:

- Laser printing & Inkjet printing
- Offset lithography
- Screen printing
- We will also look at other terms including:
- Die cutting
- Embossing
- Varnish
- Foiling
- Lamination

Digital printing

Digital printing refers to the process where the artwork sent directly to the printer in a digital format like a pdf, tiff or jpg file. An example would be your inkjet or LaserJet printer at home or work. We also get high end digital printers as well but think in terms of the printer you are familiar with at work – it works the same; you can simply click print on your computer and send it to the printer. A great example would be like when you would simply like to print 50 business cards for a client.

The advantages of digital printing are that this technique offers very high quality, its print-run is fast, you can print other colours other than the standard CMYK and you do not have to print in bulk.

Offset Lithography

Offset lithography is a printing method used for mostly mass production. The term Offset is used since the actual printing surface does not come into direct contact with the printing plate. Usually four plates (even though you can add an additional plate for spot colours, depending on the size of the printer), in in the standard CMYK colours are created - one per colour and the printed images are lasered onto these plates.

Lithography is based on the principle of oil and water do not mix. The plate goes through a process where the non-printed areas are covered with water and the printed areas are covered in an oil-based ink that then gets transferred to rollers that in return transfers the ink onto the printed surface.

Offset lithography is great for mass production, it provides consistent high-quality print, it is relatively cost effective and can also be used for shorter print runs.

Screen printing

Screen printing entails the process of stretching a piece of silk (known as the screen) over a wooden frame. The image is then transferred to the printing surface and can be used multiple times.

Screen printing is long lasting and produces vibrant colours, you can print on a variety of materials ranging from t-shirts to paper, to caps, whatever you can think of – screen printing would probably be able to print on that

You are not limited to a specific ink type, it is cost effective but takes quite a lot of time to setup, so would not be worth it to print only one or two pieces of artwork.

Letterpress

Letterpress uses a raised impression, almost like as stamp of your design to print onto the paper. You can actually feel the print. These raised images can be reused. Letterpress is used for wedding stationery, greeting cards, menus and business cards.

Letterpress creates a unique beautiful effect to your designs; however it is a tedious process that is expensive.

Other printing terms:

The following terms relate to effects that can be added AFTER the printing process

Die cutting

Die cutting is the process of using a die, which is like a stamp that cuts a cardboard or paper into the desired shape. This is mostly used for packaging, cutting the shape of the actual package and to create a sophisticated effect on other printed material like business cards and flyers. This is an expensive process, so care needs to be given.

Embossing

Embossing works similar to letter pressing but is normally added after the printing process. It simply means to raise certain elements on the printed surface to create a 3d look and feel.

Varnish

A varnish is added to create a gloss or matte overlay on parts of a design. You can think of it as a transparent spot colour

Foiling

Lastly, foiling is simply adding a gold, silver or any other shiny colour of foil on top of your design.

Conclusion

The long and short of all of this is to simply ask your printer if you are unsure about anything. Printing is not a cheap process and messing up in your pre-run is not a nice feeling – ask me, because this has happened to me before. We have looked at advanced colour theory today specifically referring to colour psychology. We have also investigated all the other colour models we get. I ended off with some of the main printing terminologies and techniques.

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